FINAL ASSIGNMENT · AI AGENT PROJECT

CodeMap AI

Understand any codebase — and what breaks when you change it.

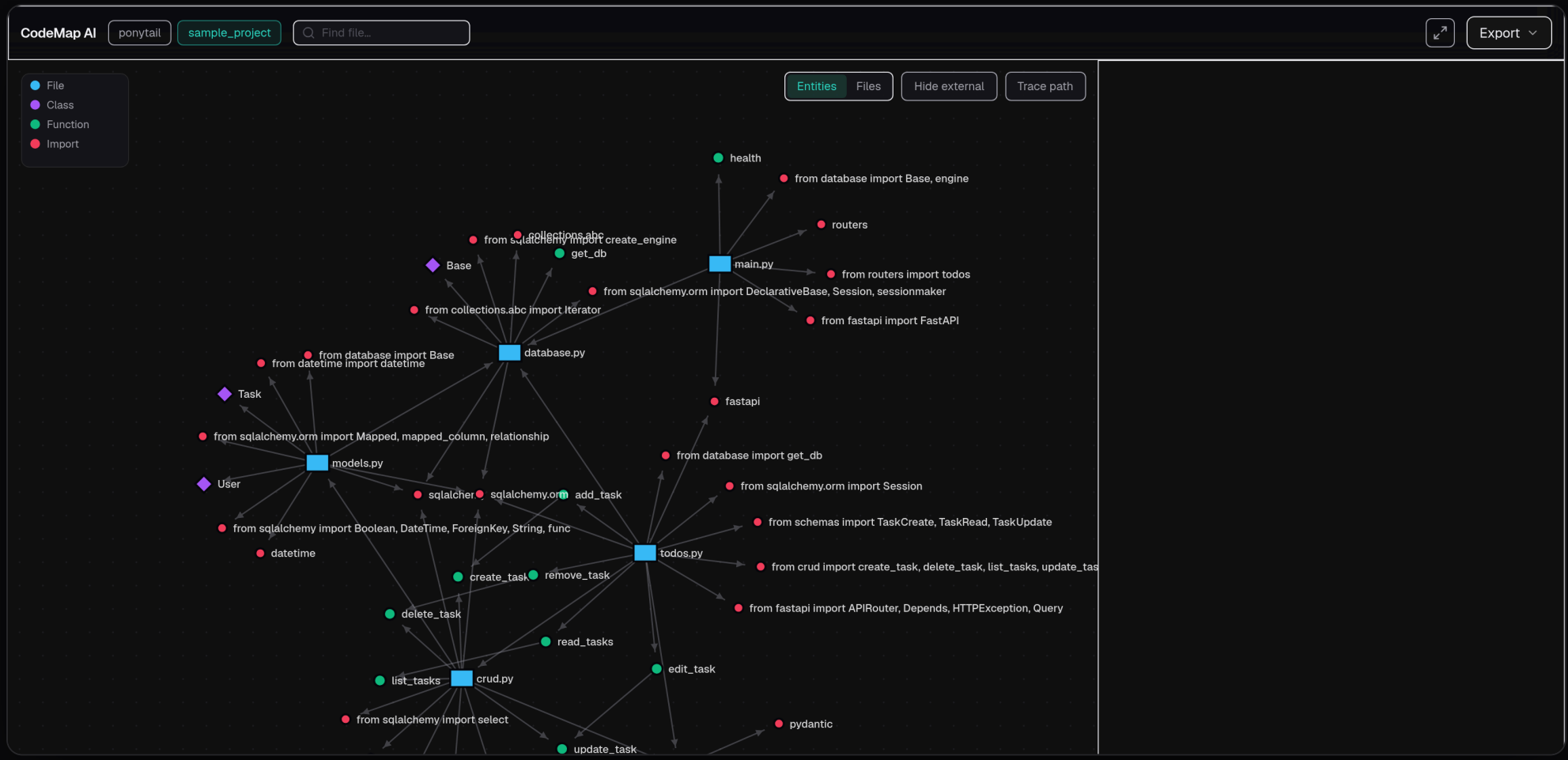
Point it at a repository. It parses the code, builds a typed dependency graph, indexes it for hybrid search, and lets you ask an agent how it works. The agent answers with citations you can click — and for change-impact questions it draws the answer on the graph.

NEXT.JS · FASTAPI · TREE-SITTER · CHROMADB · D3.JS · REACT AGENT

THE PRODUCT

A graph-first workspace

Files, classes, functions and imports as one interactive dependency graph — with filters, path tracing, and a live insights panel ranking the most depended-on files.



Typed edges
contains · import · call — so dependency metrics stay honest.

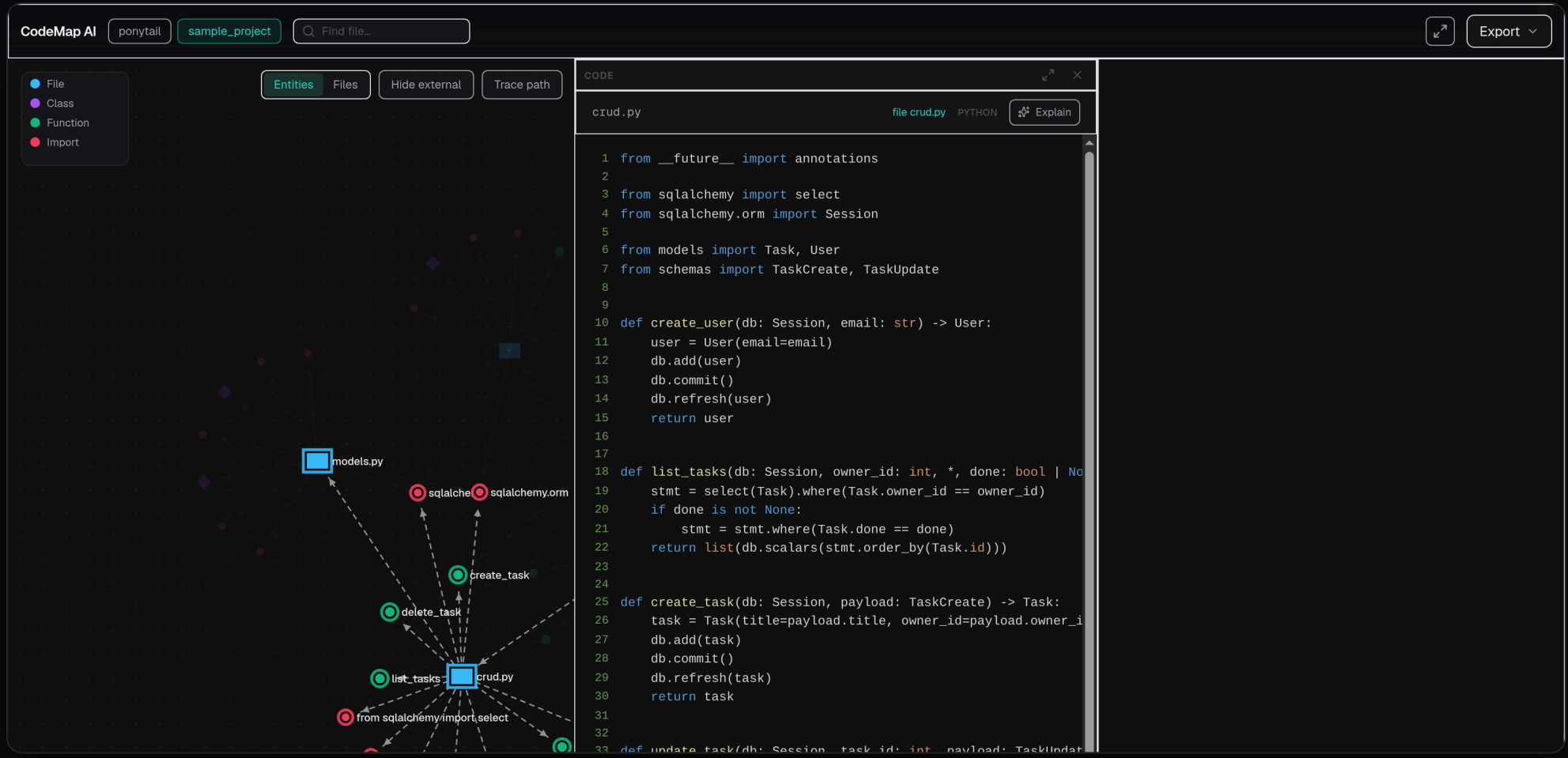
Real resolution
Relative and absolute imports resolved to actual files.

Insights
Hub ranking from in-degree, cycle detection over the dependency graph.

INTERACTION

Click a node, read its code

Selecting a node spotlights its neighbourhood and opens the source beside the graph, scrolled to the definition. The rest of the graph fades so the local structure is readable.



Focus mode
Neighbours stay lit; unrelated nodes dim to near-zero.

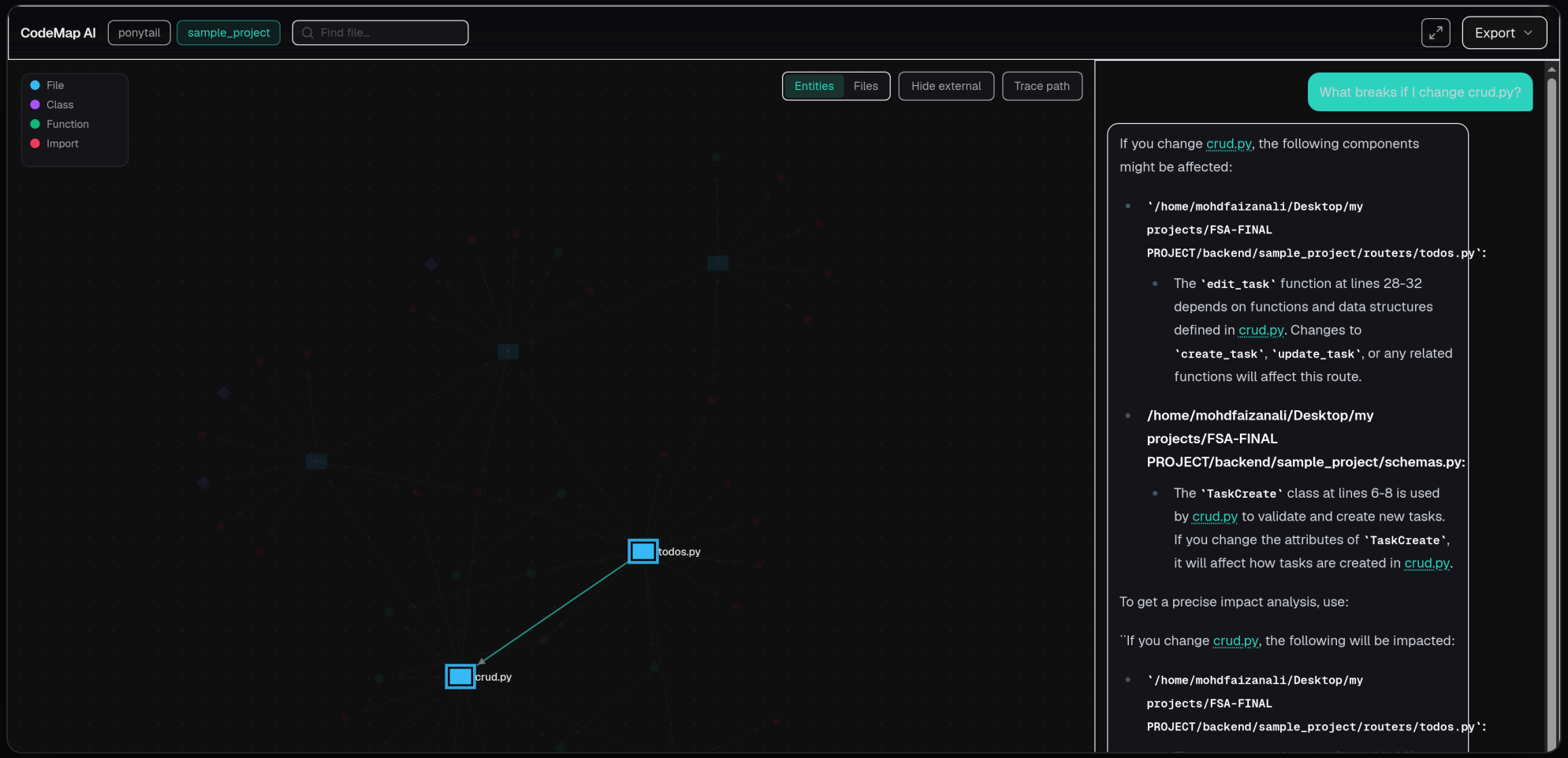
Syntax highlighted
Prism with line numbers, and the definition line marked.

Explain
Asks the agent about that exact symbol, not just the file.

THE HEADLINE FEATURE

Blast radius: what breaks if I change this?

The agent walks the reverse-dependency closure and the graph lights up exactly the affected nodes. A text search cannot answer this — it needs traversal.



The question
"What breaks if I change crud.py?"

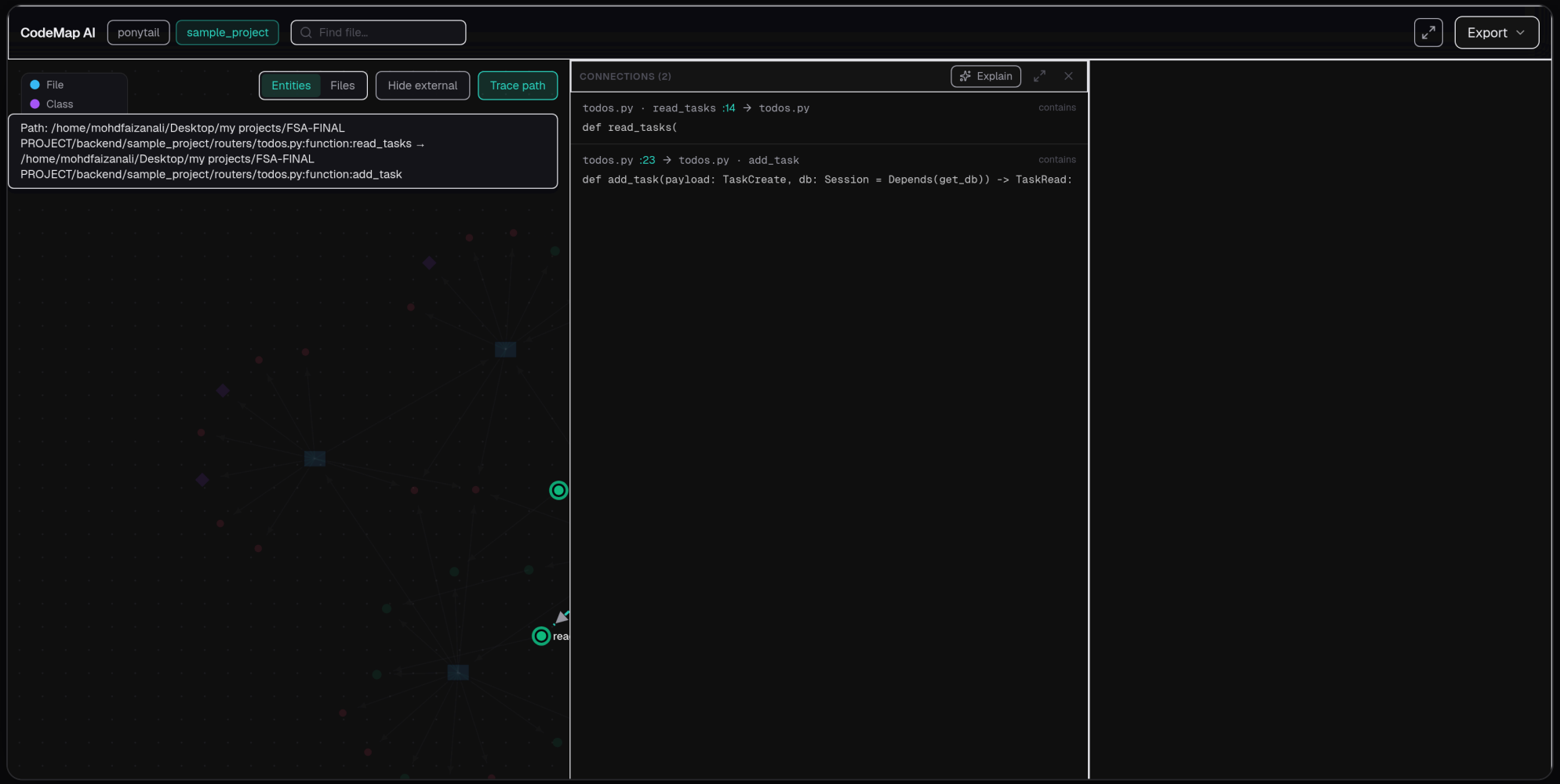
The tool
`impact_analysis(node)` — transitive dependents, ranked by hops.

The proof
2 nodes lit, 49 dimmed — the answer drawn, not asserted.

CONNECTION TRACING

How are these two connected?

Trace a path between any two nodes. Every hop shows the real source line that creates the link — and Explain narrates why they depend on each other.



Edge provenance
Each edge records the line that produced it.

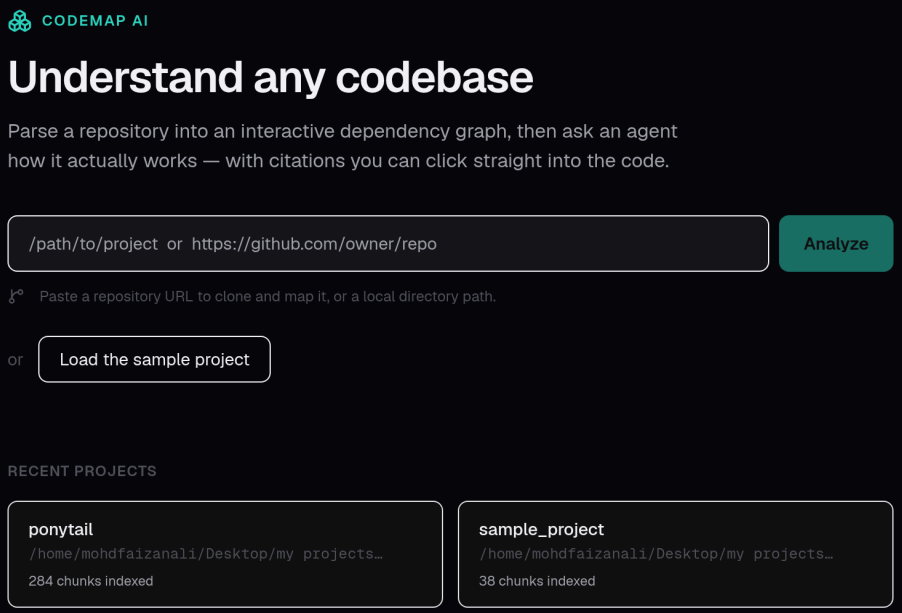
Clickable hops
Open the linking file at that exact line.

Narrated
The agent explains the relationship in plain English.

GETTING STARTED

Local path or GitHub URL

Analyse a directory on disk, or paste a repository URL and the backend clones and indexes it. Recent projects are one click away; the sample loads instantly.



Clone & index
Any git URL — validated before a subprocess runs.

Load sample
A pre-indexed FastAPI app for instant demos.

Multi-project
Isolated graphs and vector collections per project.

UNDER THE HOOD

How it works

The pipeline

- **Parse** — tree-sitter extracts functions, classes and imports from 7 languages.
- **Graph** — typed edges, resolved imports, an entity-level call graph, cycle detection.
- **Chunk** — AST-aware chunks with a context header carrying file, symbol and line range.
- **Index** — ChromaDB with Jina code embeddings.
- **Agent** — a ReAct loop over seven tools, streamed to the UI.

Why it is different

- **Hybrid retrieval** — vectors plus BM25, fused by reciprocal rank fusion, so exact identifiers work too.
- **Traversal, not paraphrase** — impact and connection questions are graph queries.
- **Typed edges** — containment excluded from metrics, so the numbers mean something.
- **Resilient by design** — token budgeting, transcript compaction, and an automatic fallback provider.

7	7	54	3	~3k
languages parsed	agent tools	backend tests	routes	tokens worst-case request